server2 retained world widget

AK retained existing widget support on 2026-10-17. `GET /ww/010000/current/2?gcode=lat,lon` is a required native supplement outside the historical 19 method/path groups. Its historical request volume is unknown. This guide covers the current legacy endpoint contract and local verification; it does not authorize activation or qualify a released iOS binary.

Request and selection

Keep `dsf` in the process's existing `SERVER2_NATIVE_FAMILIES` list. This enables the widget endpoint on the same funded world weather and air engine. Preserve the other selected families and operator-injected configuration. The widget needs no geocoder and no legacy origin. Disabling `dsf` removes this native route; it does not implement a widget fallback to legacy.

The exact retained request is:

```
GET /ww/010000/current/2?gcode=35.68,139.76
Accept-Language: ko
```

HEAD has the GET status and length with no body. A matching ETag returns 304. The successful response has no added Cache-Control policy. Missing or malformed `gcode` returns 400 locally. Unused world categories and routes are outside this supplement.

Preserve the raw contract

The current legacy route returns `timezone`, `location`, `pubDate`, `daily`, `thisTime`, `hourly`, `shortest`, `source` and `units`. It retains raw `temp_c/temp_f`, `tempMin_c/tempMax_c`, local date strings and original icon case. Hourly data begins at yesterday's local midnight and ends at most 48 hours ahead. Daily data begins yesterday. Units are metadata on this route: requesting Fahrenheit does not run the coordinate route's conversion middleware. Air fields belong on the current row only.

The active TodayWeather controller already reads raw `temp_c/temp_f` for current conditions and raw selected-unit daily values for today. Its shared expanded view also serves normalized coordinate and KMA responses. #2811 repairs that view: dotted raw dates use `tempMin_c/tempMax_c` or `tempMin_f/tempMax_f`; compact `YYYYMMDD` dates keep the already requested-unit `tmn/tmx`. Preserve the six-day limit, icons, localized first three day labels and integer display. Short or non-string dates are guarded before substring access in the renderer; future date labels become `--/--`. This is not a general malformed KMA-input recovery guarantee.

The widget target no longer links the unused Cordova archive. The TodayWeather app target retains Cordova. TodayAir and Apple Watch code are unchanged.

Check the source-built display

Use the app's temperature preference and a saved overseas city, then open the expanded six-day forecast. In the owned UIKit replay, the actual storyboard and three source-built widget objects show London at 16.7 degrees C or 62 degrees F, today 10/22 degrees C or 50/73 degrees F, and six corresponding daily rows. The selected actual capture shows the Fahrenheit run; the verification record gives source and command identities.

The regression replay includes raw C/F, normalized coordinate C/F, KMA C/F, a short date, zero and negative values, a year boundary and the six-row limit. Its separate functional replay passes the retained native `/ww` response through `processGlobalAddress` and the real controller. An intercepting `NSURLProtocol` accepts only the exact synthetic loopback request and rejects every other HTTP(S) request. No public provider, production key, AWS operation or live push is involved.

Apple removed legacy Today View extensions in iOS 18. The iOS 26.5 Simulator therefore validates an owned UIKit container, not Today/Springboard registration. The real widget target builds, but neither an App Store IPA nor a deployed widget was exercised. Modern WidgetKit activation remains the separate #2605 scope. This repair does not establish that an existing released widget is available on current iOS.

Failure and recovery

An initial provider authentication failure remains HTTP 401. Requests during the provider-down window return HTTP 503 with `EWEATHERUNAVAILABLE`, `retryAt`, Retry-After and no-store. Generic provider failure remains HTTP 500. Native weather acquisition, quota ownership, S3 publication, stale/failure handling and air deadlines are shared with the existing world implementation. The new path does not reset budgets or add retries.

Warm requests reuse the cell. After a process restart, S3 remains authoritative for weather and air. Local verification disables the owned provider peers after the first successful publication, then checks that a new process serves identical bytes with no additional weather, air, geocoder or legacy calls. Missing production state must terminate natively, not depend on legacy.

Repeat the owned verification

Build the release `server2` binary and `loopback_server` example under `server2/`, then run:

```
python3 server2/tools/widget_smoke.py --binary <release/server2> --output <new-evidence-directory>
```

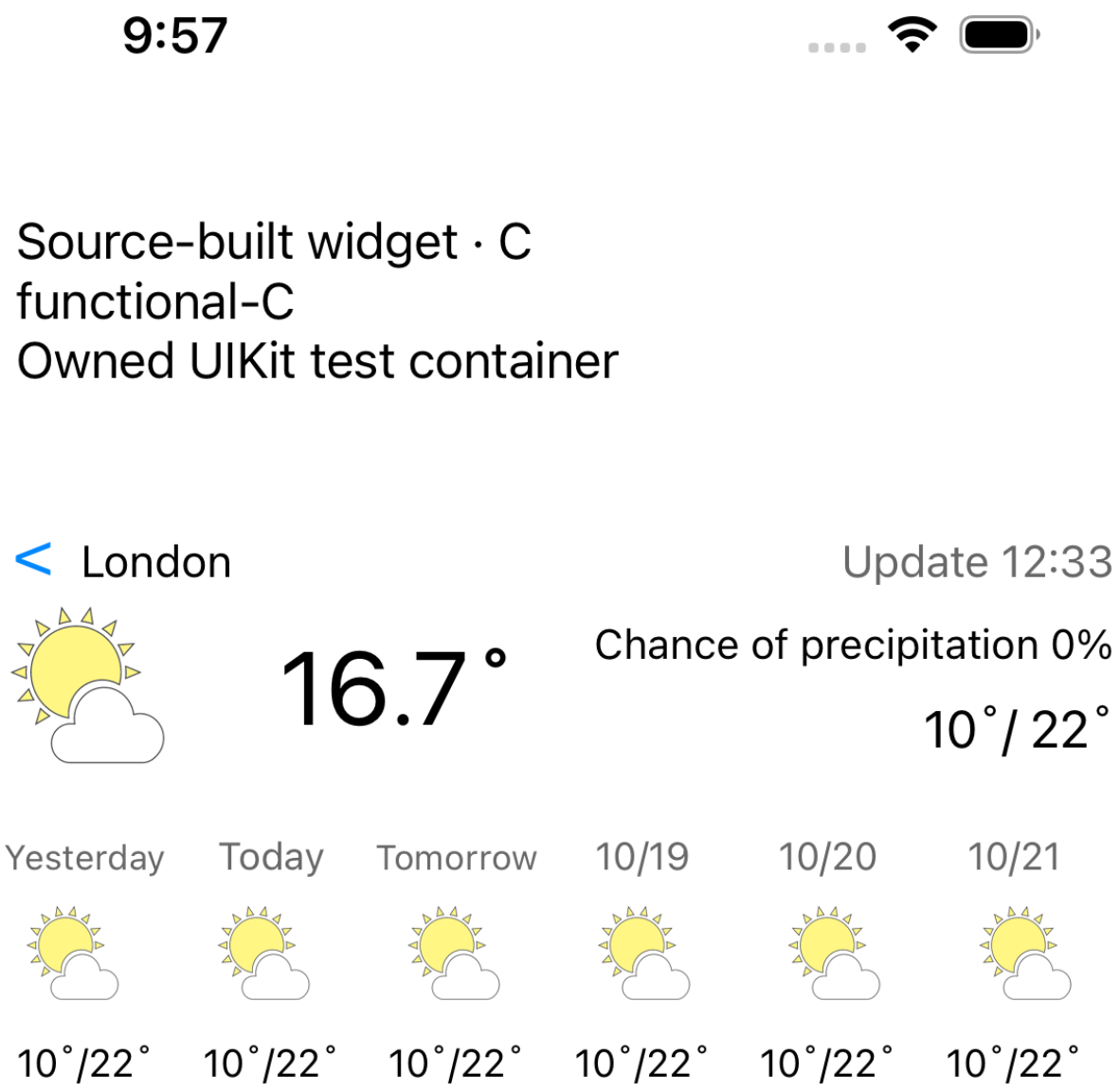
The smoke starts only owned loopback S3 and synthetic recorded-format weather/air peers, uses dummy credentials in an empty working directory, writes a report and diagnostic logs, and retires its processes. It makes no public provider, AWS, push or device call. The report includes executable hashes, cold/warm results, conditional and error checks, restore results and call counts. Device-sends zero describes this isolated no-send setup; it is not a production device counter.

The selected verification record binds the actual release run. Cutover preparation still requires production revision parity, seven-day paired shadow, host/listener/capacity and quota measurements, S3/IAM/IMDSv2 readiness, final state/rainfall handover, an eligible complete previous server2 bundle, and separately scoped approval. Local PASS does not close #2704/#2705 or make production eligible.

Select Celsius and expand the forecast

Keep the existing saved city and select Celsius. Open the expanded forecast. The actual controller is driven by the retained native WW response in an owned test container.

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Actual Celsius replay: London, 16.7 degrees current, 10/22 daily; six day labels and icons.

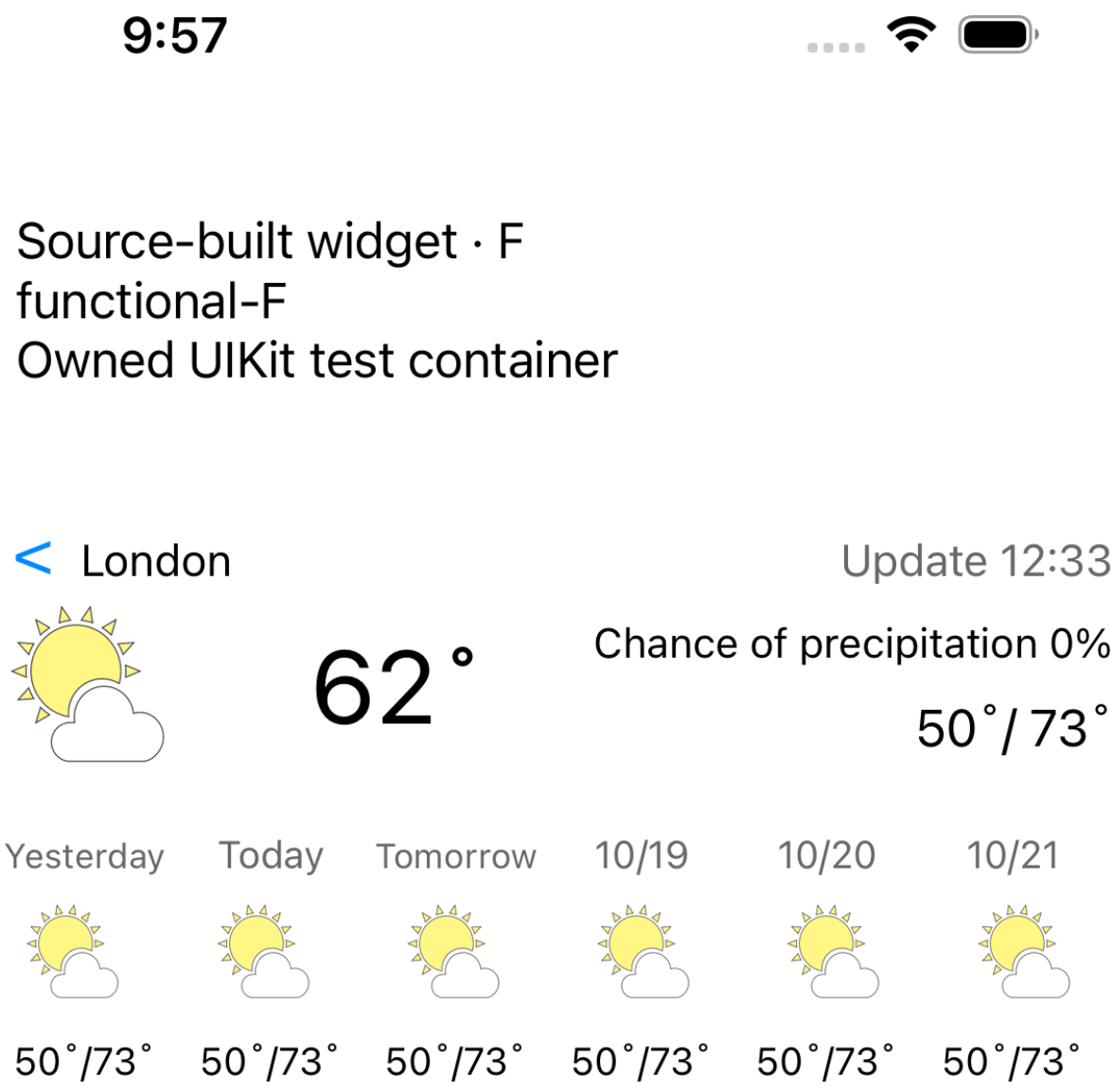
Expected: Both compact and expanded values agree with raw Celsius fields. Normalized coordinate and KMA C/F regression cases also pass.

Unmodified Simulator capture, top 1350-pixel viewport shown; blank lower test-container area omitted from this page.

Select Fahrenheit and check all six days

Select Fahrenheit and expand the same city. The replay isolates HTTP to an exact loopback request; other HTTP(S) requests are rejected.

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Actual Fahrenheit replay: London, 62 degrees current, 50/73 daily. Dates and six row limit are retained.

Expected: Fahrenheit uses raw F values without converting normalized values again. No modern widget activation is claimed; Apple removed Today extensions in iOS 18.

Unmodified Simulator capture, top 1350-pixel viewport shown; blank lower test-container area omitted from this page.